

SETIAWAN MUHAMMAD

SOFTWARE ENGINEER · DEVOPS & CYBERSECURITY SPECIALIST (SOLO L1/L2/L3 OPERATOR)

Surabaya, Indonesia · Remote Ready

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EXECUTIVE SUMMARY

Defensive & Offensive Security · Production Infrastructure · Distributed Systems

Disciplined Software, DevOps, and Cybersecurity Engineer with hands-on production ownership spanning solo L1/L2/L3 security operations, high-availability Linux infrastructure, and high-concurrency software architectures. Proven track record at **DoubleLogic** solely orchestrating the entire security lifecycle across 7+ production Linux servers (incident response, RCE backdoor eradication, WAF virtual patching, and AI Red Teaming) alongside zero-downtime database disaster recovery pipelines (MySQL & MongoDB to Cloudflare R2). Adept in high-throughput backend engineering (Golang, Python, PostgreSQL concurrency controls) and academic mentorship as a Telkom University CS Teaching Assistant.

TECHNICAL COMPETENCIES

Core Engineering & Operational Stack

Cybersecurity & SOC:	Solo L1/L2/L3 Operations, Incident Response & Memory Forensics (IOC extraction), Web Application Penetration Testing, AI Red Teaming & LLM Safety (Prompt Injection) , Cloudflare WAF Tuning & Virtual Patching, STRIDE Threat Modeling, OWASP Top 10, IAM Auditing, Linux Kernel/SSH Hardening.
DevOps & Infrastructure:	Docker & Docker Compose, Linux Administration (Ubuntu/Debian), Nginx Load Balancing & Reverse Proxy, Cloudflare R2 / Exabytes Encrypted DR (RPO < 1h), Grafana & Prometheus Telemetry, Timezone-aware GPU Auto-Scheduler, CI/CD (GitHub Actions), Bash Automation.
Backend & Languages:	Golang (Gin, Fiber, Concurrency, Mutex, pgx), Python (FastAPI, Flask, Asyncio, Scikit-Learn), TypeScript/JavaScript (Node.js, Express, Next.js, React 19), PHP (Laravel 11/12, Filament v3), Java OOP, C++, Dart (Flutter).
Databases & Systems:	PostgreSQL (Pessimistic Locking `SELECT ... FOR UPDATE`, Composite Indexing), Redis 7 (Pub/Sub, Distributed Locking, TTL Queues), MySQL 8, MongoDB, ChromaDB (Vector Store), SQLite.
Software Architecture:	Clean Architecture, Hexagonal / Ports & Adapters, Microservices, Event-Driven Streaming, Race Condition Mitigation, RESTful APIs, WebSockets, RAG (Retrieval-Augmented Generation).

PROFESSIONAL WORK EXPERIENCE

Production Infrastructure & Security Operations

Cyber Security Operations & Red Team Engineer — DoubleLogic

SOLO L1/L2/L3 OPERATOR

Post-Promotion - Present

FULL-TIME REMOTE

Solely led defensive and offensive security operations across 7+ production Linux servers, safeguarding proprietary AI services, databases, and client web endpoints without secondary staff.

SINGLE-OPERATOR SECURITY LIFECYCLE EXECUTION (SOLO L1 · L2 · L3)

LEVEL 1 · TRIAGE & TELEMETRY

- Real-time alert triage & log ingestion across 7 production host servers.
- Monitored UFW/iptables firewall event logs & authentication anomalies.
- Instant incident routing via custom MS Teams webhook pipelines.

LEVEL 2 · FORENSICS & RESPONSE

- Investigated & eradicated active RCE backdoor in DrTKola/YEG (Server 139).
- Led forensic root-cause analysis on PMCare Ameyo attack vector.
- Emergency post-incident malware & rootkit scans across all 7 servers.

LEVEL 3 · RED TEAM & AI SAFETY

- Engineered Cloudflare WAF tuning & virtual patching against zero-days.
- Simulated internal lateral movement, privilege escalation, & BOLA audits.
- Conducted AI Red Teaming: prompt injection & LLM agent jailbreak defense.

- Infrastructure Hardening:** Audited IAM access, rotated SSH keys, eliminated weak passwords, and enforced strict PAM/UFW rules across all nodes.
- Tooling & Automation:** Authored custom Python security scripts for automated endpoint recon, port auditing, and database access control verification.

DevOps & Infrastructure Engineer — DoubleLogic

INITIAL TENURE

Remote · Cloud Infrastructure

- Server Fleet Management:** Administered dozens of Ubuntu Linux servers across Production, Staging, and GPU clusters, maintaining 99.9%+ availability.
- Cost Optimization:** Engineered a timezone-aware GPU auto-scheduler script, slashing idle cloud compute overhead substantially during off-peak hours.
- Disaster Recovery Automation:** Built encrypted database backup pipelines (MySQL & MongoDB) replicated to Cloudflare R2 and Exabytes with verified automated restore drills (RPO < 1 hour via SSH tunneling).
- Observability & Middleware:** Deployed centralized Grafana monitoring stacks (CPU, RAM, Disk, Nginx traffic) with Teams alerting; authored Python Flask/Gunicorn webhook services for high-throughput email and attachment ingest.

PROFESSIONAL WORK EXPERIENCE (CONTINUED)

Client Engagements · Academic Leadership

Software Engineer & Security Researcher — Independent Consultancy / Client Engagements (Upwork) CONTRACT 2023 – Present

- **API Integration & Production Bug Fixing:** Refactored mission-critical WhatsApp integrations using Evolution API on production servers, resolving race conditions, connection timeouts, and memory leaks.
- **Helpdesk & Ticketing Infrastructure:** Deployed and customized UVdesk enterprise platforms including multi-mailbox integration and automated workflow routing.
- **Wellness Booking Engine:** Architected end-to-end booking and scheduling web platforms for wellness businesses with automated transactional billing and notifications.
- **Vulnerability Research:** Identified, documented, and responsibly patched critical vulnerabilities (SQLi, Stored XSS, BOLA/IDOR) on client web architectures.

Computer Science Laboratory & Teaching Assistant — Telkom University

Sep 2023 – Present · Surabaya, ID

ACADEMIC APPOINTMENT

- **Technical Instruction:** Mentored 100+ undergraduate engineering students across 8 subjects: Algorithms & Programming (Golang), Machine Learning (Python), Information System Security (Kali Linux), Cloud Computing (Azure), Data Structures (C++), Operating Systems (Linux), OOP (Java), and Computer Networks (Wireshark).
- **Lab & Code Governance:** Maintained Cyber Security & Cloud Computing lab environments; conducted weekly code reviews on 300+ submissions per semester, enforcing Clean Architecture, SOLID principles, and Big-O efficiency.

FEATURED SIGNATURE PROJECTS · DISTRIBUTED SYSTEMS & CONCURRENCYHigh-Performance Core Architectures

ForgeFlow

Go · Postgres · Redis · Docker

Distributed Workflow Orchestration Platform

Engineered a self-hosted, distributed task execution engine in Go with PostgreSQL-backed state persistence, Redis Pub/Sub coordination, and graceful worker shutdown. Guaranteed at-least-once task delivery under worker node failures.

NOVA AI Coding Agent

Go · Bubble Tea · Ollama · AST

Autonomous Zero-Leak Terminal Coding Agent

Engineered model-agnostic coding agent in Go using Hexagonal Architecture. Features interactive TUI (Bubble Tea), AST project workspace indexing, deterministic capability routing with JSON Schema validation, and zero telemetry leaks.

AURIX v11.5b Engine

Go · Python · Redis 7 · Next.js

Low-Latency Algorithmic Trading & Quant Engine

Distributed quant trading platform decoupled across 8 Docker containers. Features sub-millisecond WebSocket market bus in Go, Layer-4 capital protection with Redis `EXPIREAT` TTL caps (-2.0%), and forward-validated 71.4% win rate.

SafeTicket & FlashSale

Go · PostgreSQL · K6 · Nginx

High-Concurrency Booking & Transaction Engine

Engineered high-throughput flash-sale backend in Go comparing concurrency mechanisms. Enforced PostgreSQL Pessimistic Locking (`SELECT ... FOR UPDATE`), guaranteeing zero overselling under multi-thousand concurrent virtual user stress tests.

FEATURED SIGNATURE PROJECTS · SECURITY, AI & ENTERPRISE PLATFORMS

Offensive Security · Enterprise SaaS · Computer Vision

Stockbit AppSec Simulation

Go · STRIDE · OWASP · Docker

Fintech Secure SDLC & Threat Mitigation Prototype

End-to-end Secure SDLC implementation executing STRIDE threat modeling, defensive Clean Architecture, contextual RBAC, and automated security unit test suites mitigating OWASP Top 10 vulnerabilities in financial transactions.

BelanjaKu

Next.js 15 · Express · PostgreSQL · Zustand

Production Decoupled E-Commerce Ecosystem

Production-grade e-commerce platform with decoupled architecture. Implemented persistent client-server cart sync via Zustand, structured prompt product generation, JWT auth, and automated test suites.

The Money Maker

Go · Python · Isolation Forest · Redis

Cross-Exchange Triangular Arbitrage Engine

High-frequency arbitrage engine pairing concurrent Go WebSocket stream handlers with Python Scikit-Learn Isolation Forest for real-time fakeout liquidity and anomaly filtering across Binance and Bybit.

Pentara (Pentest Nusantara)

Laravel 11 · PHP 8.2 · Python · MySQL

Multi-Tenant B2B SaaS Bug Bounty Platform

Architected multi-tenant vulnerability disclosure SaaS with granular RBAC (Admin, Enterprise, Researcher), automated CVSS 3.1 scoring engine, encrypted PoC file handling, and executive PDF audit report generation.

Veritas (Efficiency Beast)

Python · FastAPI · ChromaDB · RAG

Zero-Hallucination Enterprise RAG Platform

Local-first document Q&A platform with ChromaDB vector store. Features deterministic chunking, mathematical source attribution guardrails eliminating hallucinations, and sub-200ms document retrieval.

VisionGuard YOLO11

Python · YOLO11 · PyTorch · Computer Vision

Real-Time Object Detection & Logo Identifier

Fine-tuned YOLO11 model detecting evasive illicit site branding and logos across diverse screenshot layouts in real-time with sub-20ms inference and high precision mAP.

EDUCATION

Bachelor of Science in Informatics

Expected 2027

Telkom University · Surabaya, ID

Current Standing: Semester 6/7 · **Cumulative GPA:** 3.46 / 4.00

Relevant Focus: Distributed Systems, Computer Networks & Protocols, Database Systems, Operating Systems, Information Security, Clean OOP.

CERTIFICATIONS & CREDENTIALS

- **Code Generation & Optimization with IBM Granite** — IBM (2024)
- **Ethical Hacking & Penetration Testing from Scratch** — Udemy (2024)
- **Mastering Linux & System Administration** — Udemy (2023)
- **Backend Development with JavaScript** — Dicoding (2023)
- **Fundamentals of Artificial Intelligence** — Dicoding (2024)
- **Android Application Development** — Dicoding (2023)

LANGUAGES & OPERATIONAL READINESS

Indonesian: Native / Bilingual Proficiency · **English:** Professional Working Proficiency (Technical Architecture, Audits, RFCs, & Global Client Delivery)

Availability: Full-Time / Contract Remote · WFA Ready · Immediate Operational Deployment